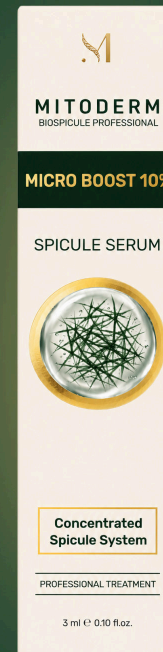


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# BIOSPICULE - BIOMICRONEEDLING

4TH GENERATION



# BIOSPICULE™ – What is Spongilla?

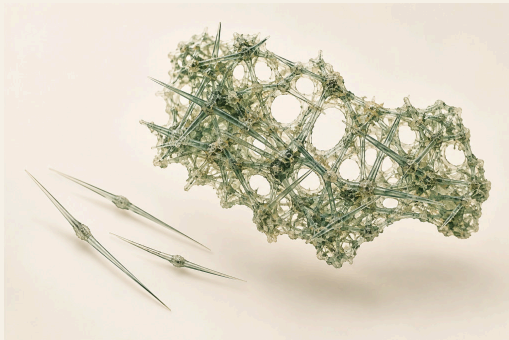
**Spongilla** is a genus of freshwater sponges belonging to the phylum Porifera. Unlike the vast majority of sponges, which live in the sea, this genus is found in lakes, streams, and other freshwater environments. Despite its plant-like appearance, it is a simple aquatic animal with no nervous system, organs, or muscles.

The sponge has a skeleton that supports its structure and consists of two main components:

Organic fibres, mainly a collagen-like protein called spongin.

Tiny mineral needles made of silica, called spicules.

**Spongilla** is the sponge; **spicules** are the microscopic needles that form its skeleton.



## Millions of micro-needles penetrate rapidly and reach deep into the skin

Millions of micro-needles penetrate the skin to a depth of 80–300 microns, reaching the epidermal layers. This mechanical stimulation activates **microcirculation**, accelerates cellular **metabolism**, and supports the skin **renewal** cycle.

Under the microscope, tiny spicules can be seen within the skin.

They may remain for up to 72 hours, providing continued mechanical **stimulation** and supporting collagen production, epidermal repair, cellular metabolism, and skin renewal.

### Indications

Anti-ageing | Wrinkles and fine lines | Enlarged pores | Firming | Brightening | Uneven skin tone | Post-inflammatory hyperpigmentation (PIH) | Senile lentigo | Scars | Post-acne scars | Mild to moderate acne | Thin, devitalised skin | Dull, sallow skin | Renewal and nourishment | Face | Neck | Décolleté | Hands | Back

**Micro-needle treatment can serve as an alternative to traditional microneedling.**

# BIOSPICULE™ – The Fourth Generation of Spicule Technology

## What Makes Us Different?

Spongilla-based technologies depend on several key factors:

The purification level of the spicules.

The size of the spicules and the ratio between different lengths.

Their precise concentration in each product.

**Traditional Spongilla powder is a single, non-uniform mixture** containing the entire raw material.

### **Our approach:**

Fourth-generation BIOSPICULE™ technology uses **four types** of Spongilla. Through an advanced Korean processing method, unwanted residues are removed, while the microscopic structures are purified and sorted according to their characteristics and size.

This allows us to control the structure and characteristics of the spicules with greater precision. **Every bio-needle matters.**

## Why Does Purification Change Everything?

Rather than focusing only on the quantity of Spongilla, our technology also considers the architecture of the spicules – their length, the balance between short and long structures, and how this combination influences the distribution of mechanical stimulation across different layers of the epidermis.

This is where a natural raw material becomes a precise cosmetic technology.

Under the microscope, each spicule has a porous structure with an elongated microscopic inner cavity, almost like a miniature cannelloni. This unique architecture makes the spicule more than a mechanical element: it becomes part of a controlled delivery system.

By controlling different spicule lengths within the formula, BIOSPICULE™ combines mechanical stimulation with targeted cosmetic delivery. This structure forms the basis of the Hybrid Spicule Delivery System.

# Two Parallel Mechanisms of Action

## Skin Renewal – Resurfacing

During massage, the spicules penetrate the stratum corneum, creating controlled mechanical stimulation. This supports the gradual removal of dead skin cells, improves skin texture and appearance, and prepares the skin for active ingredients applied later in the treatment.

The mechanical action of the spicules helps improve the appearance of roughness, enlarged pores, dullness, and uneven texture.

## Delivery – Cosmetic Delivery System

The spicules create temporary microscopic pathways in the outer layers of the skin, supporting more effective delivery of active ingredients applied during the treatment.

The aim is not only to create a peeling effect, but to transform the renewal stage into a precise cosmetic delivery pathway — BIOMICRONEEDLING.

Biological micro-spicules may remain in the skin for up to 72 hours, providing continued mechanical stimulation.



**Spicule → Resurfacing → Delivery**

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# MICRO BOOST 10%

10%

BIOSPICULE™

Professional In-Clinic Treatment

The most **concentrated professional** serum in the BIOSPICULE™ range. Designed exclusively for in-clinic use, it comes in a 3 ml syringe, allowing the aesthetician to control the amount of product, massage duration, and treatment intensity according to the skin diagnosis.

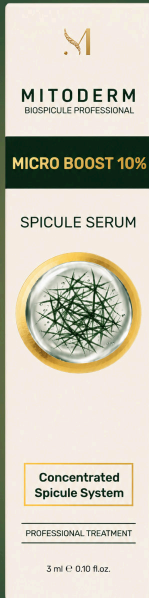
The formula contains 10% Hydrolyzed Sponge and combines two groups of spicules:

40% Mini Spicules, approximately 80–100 microns in length.

60% Long Spicules, approximately 300–360 microns in length.

The combination of short and long spicules creates different levels of mechanical stimulation. Short spicules act mainly in the outer layers of the epidermis, while longer spicules provide a more intense stimulus and reach deeper layers of the skin.

The aesthetician therefore works not merely with a peel, but with a BIOMICRONEEDLING delivery system – a professional approach combining skin **renewal** with the **delivery** of active ingredients.



## Who Is the Treatment For?

Subject to professional assessment:

Anti-ageing | AGING | Dull, tired skin lacking vitality | Rough or uneven texture | Enlarged pores | Uneven skin tone | Post-acne scars | Surgical scars | Pigmentation and sun spots | Post-inflammatory pigmentation (PIH) | Wrinkles, fine lines and signs of ageing | Loss of elasticity and firmness | Smokers' skin | Mild to moderate acne | Firming | Improvement in overall skin quality

Due to the high concentration of spicules, a characteristic tingling sensation may occur during treatment and in the days that follow, particularly upon contact with the skin. Temporary redness, dryness, tightness, or varying degrees of peeling may also occur.

## Professional In-Clinic Combinations

The structure of the spicules and the massage technique create temporary microscopic channels, allowing the treatment to be combined with:

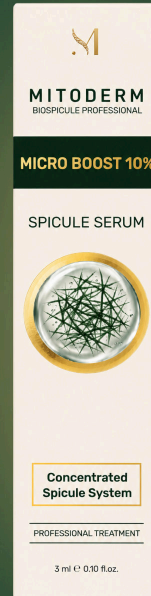
V-TECH or Longevity Serum

**Treatment Finish:**

EXOCELL MASK

DERMA RECOVERY CREAM – also mandatory for home use to support the epidermal barrier and reduce dryness and tightness

Sunscreen



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# BIOSPICULE<sup>TM</sup> series

CELL RENEW 1.5% – home-care serum for renewal and brightening

CELLULAR AGE DEFENSE 2.5% – advanced serum for firming and anti-ageing

DERMA RECOVERY CREAM – repairing cream for soothing and skin barrier support



# CELL RENEW 1.5%

1.5%

## BIOSPICULE™ SERUM

At-Home Serum for Skin Renewal and Brightening

A spicule serum designed to gradually introduce the skin to an advanced renewal routine, with a focus on achieving a more radiant, smooth, and even-looking complexion.

The formula combines a moderate concentration of spicules with ingredients that promote hydration and freshness while helping improve skin texture and tone. During massage, a gentle, characteristic tingling sensation may occur due to the mechanical contact of the spicules with the outer layers of the skin.

This is not an ordinary at-home peel. The spicules provide moderate mechanical stimulation and support the renewal processes of the skin's surface, while the active ingredients in the formula provide complementary support for skin renewal, rebuilding, radiance, and a more even appearance.

## Indications

Suitable for all skin types | Fresh, radiant-looking skin | Dull and tired-looking skin | Improved skin texture | Enlarged pores | Younger-looking skin | Improvement of uneven skin tone | Hyperpigmentation | Post-inflammatory marks | Skin renewal and brightening | Anti-ageing | Improvement in the appearance of wrinkles and fine lines | Firming

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Advanced Biospicule Technology  
30 ml e 1.0 fl.oz.

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# CELL RENEW 1.5%

## BIOSPICULE™ SERUM

Home-Care Serum for Skin Renewal and Brightening

### Key Active Ingredients

#### Hydrolyzed Sponge – Spicules at a Concentration of 1.5%

Microscopic needle-like structures that provide controlled cosmetic mechanical stimulation to the skin. They support the gradual removal of dead skin cells and prepare the outer layers of the skin to receive the formula.

#### Niacinamide

A multifunctional ingredient that supports the skin's moisture barrier and helps improve the appearance of uneven skin tone and enlarged pores. It helps leave the skin looking calm, even, and refreshed.

### Galactomyces Ferment Filtrate

A yeast-fermented filtrate widely used in Korean skincare. It supports a smooth, vital, and radiant skin appearance, helping improve the overall look and feel of the skin.

#### Sodium DNA

Purified DNA fragments derived from rice. They help retain moisture, support the epidermal barrier, and contribute to a smoother, more supple, and revitalized skin appearance.

## Salicylic Acid + Citric Acid

A combination of salicylic acid and citric acid that helps renew the skin's surface, improve rough texture, reduce the appearance of enlarged pores, and promote a brighter, more even-looking complexion.

## Sodium Hyaluronate, Glycerin & Betaine

A combination of moisturizing ingredients that helps attract and retain water in the outer layers of the skin, reduce the feeling of dryness, and leave the skin feeling softer and more supple.

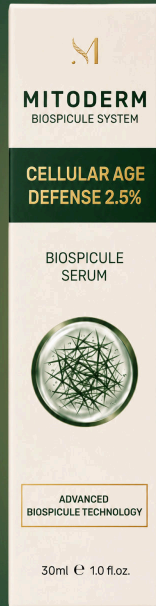
## How to Use

For daily use on clean, dry skin. Massage gently until absorbed. A mild, characteristic tingling sensation may occur during application. There is no need to apply firm pressure or rub the skin.

After applying the serum, follow with Derma Recovery Cream. During the day, always apply sunscreen to complete the routine.



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Advanced Biospicule Technology  
30 ml e 1.0 fl.oz.

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# CELLULAR AGE DEFENSE 2.5%

## BIOSPICULE™ SERUM

Advanced Serum for Firming and Anti-Ageing

Developed for skin that requires a more intensive care programme, combining a higher concentration of spicules with Bakuchiol, five peptides, and moisture-supporting ingredients.

Compared with CELL RENEW 1.5%, this serum provides a more intensive and noticeable treatment experience, with a focus on the appearance of signs of ageing and fine lines.

### Suitable For:

All skin types | Improved appearance of wrinkles and fine lines | Anti-ageing | Improved elasticity and overall skin quality | Tired, dull-looking skin lacking vitality | Uneven or rough skin texture | Reduced appearance of enlarged pores | Firming and brightening | Improved appearance of post-acne scars | Mild to moderate acne.

**2.5%**

# CELLULAR AGE DEFENSE 2.5%

## BIOSPICULE™ SERUM

Advanced Serum for Firming and Anti-Ageing

### Key Active Ingredients

#### Hydrolyzed Sponge – 2.5% Spicule Concentration

The spicules support skin renewal and prepare the outer layers of the skin to receive the formula's active ingredients.

#### Bakuchiol

A plant-derived ingredient with retinol-like properties. Helps improve the appearance of fine lines, skin texture, and tone, supporting a smoother, firmer, and more even-looking complexion.

### Five Peptides:

Acetyl Hexapeptide-8 | Acetyl Tetrapeptide-5 | Palmitoyl Tetrapeptide-7 | Palmitoyl Tripeptide-5 | Carnosine  
Help improve the appearance of firmness and elasticity, support skin hydration and freshness, and improve the appearance of wrinkles and fine lines.

### Hydrolyzed Collagen Extract

Helps retain moisture and leaves the skin feeling supple and refreshed.

### Adenosine

A well-known anti-ageing ingredient widely used in Korean cosmetics, helping improve the appearance of fine lines.



## Yeast Beta-Glucan

Yeast-derived beta-glucan that helps soothe the skin, retain moisture, and support the skin barrier.

## Akebia Quinata and Camellia Japonica Extracts

Plant extracts that support a healthy, vital, and radiant skin appearance while providing antioxidant support to the formula.

## Directions for Use

For daily use. Apply to clean, dry skin and massage gently. A noticeable tingling sensation may occur, particularly upon contact with the skin. Do not rub vigorously.

After applying the serum, follow with Derma Recovery Cream to support the skin's moisture barrier and maintain a comfortable, hydrated feel. Sunscreen must be applied during the day.

For skin that is not yet accustomed to the product, a gradual transition from CELL RENEW 1.5% to CELLULAR AGE DEFENSE 2.5% is recommended, according to individual skin tolerance.

# DERMA RECOVERY CREAM

## SNAIL COLLAGEN

Restorative Cream for Soothing and Skin Barrier Support

A restorative cream that provides hydration and nourishment for daily use. Designed as a complementary cream for professional treatment protocols and daily skincare routines, particularly after the use of products containing spicules.

**The cream itself contains no spicules, and its role is complementary:** it provides hydration, softness, and comfort while supporting the epidermal barrier during the skin renewal process.

## Indications

Completion of the professional V-TECH SYSTEM protocol | After an invasive in-clinic procedure | Completion of a professional in-clinic treatment with Micro BOOST 10% | After a home-care routine from the BIOSPICULE™ range, including CELL RENEW 1.5% and CELLULAR AGE DEFENSE 2.5%

## Who Is Derma Recovery For?

Provides hydration for all skin types | General hydration after an in-clinic treatment | Hydration for dry or dehydrated skin | Hydration for skin requiring additional support after a professional spicule treatment | Skin with an irritated, rough, or flaky appearance | Skincare routines requiring hydration, nourishment, and soothing

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Daily nourishing collagen cream for  
healthy-looking glow  
50 ml e 1.69 fl.oz.

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# DERMA RECOVERY CREAM

## SNAIL COLLAGEN

Restorative Cream for Soothing and Skin Barrier Support

### Key Active Ingredients

#### Panthenol

**Provitamin B5** that retains moisture, relieves dryness and discomfort, and promotes softness and suppleness.

#### Madecassoside

Derived from *Centella asiatica*, it soothes, strengthens the moisture barrier, and provides antioxidant benefits.

#### Allantoin

Soothes and softens while reducing roughness and improving texture after cosmetic renewal procedures.

#### Shea Butter

Rich in essential lipids, it reduces moisture loss and leaves the complexion soft, nourished, and supple.

#### Niacinamide

Strengthens the skin barrier, improves moisture retention, and promotes a calm, even, healthy-looking complexion.

#### Hydrolyzed Collagen

Forms a moisture-retaining layer that enhances smoothness, suppleness, and a replenished feel.

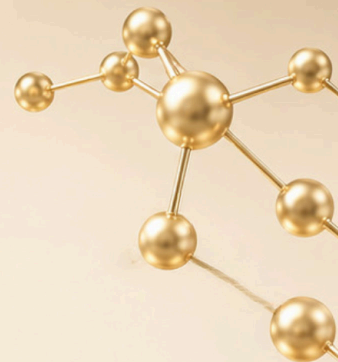
#### Snail Secretion Filtrate – Purified Polysaccharides from Snail Mucin

Binds and retains water, providing lasting hydration, softness, comfort, and a refreshed appearance.



Where Science Meets Beauty

MITODERM develops advanced professional solutions for the aesthetics industry, combining scientific expertise, innovation, and cutting-edge technology. Our extensive experience enables us to create products and protocols that adhere to the highest standards of quality, efficacy, and safety



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